

# **STA 35C: Statistical Data Science III**

## **Lecture 16: Linear Model Selection – Regularization Methods**

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# Announcements

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## Homework 5 is posted

- Please review the problems early and ask questions as needed.
- Submit a single PDF on time and follow the submission instructions.

## Office hours today

- Instructor office hours: 3:30–4:30 PM, MSB 4220.
- You are welcome to bring any questions about Homework 5 or recent lecture material.

# Agenda

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- **Last time:** Model selection via subset selection
  - Best subset selection: identify relevant predictors among many
  - Stepwise selection: A computationally cheaper greedy alternative
- **Today:** Regularization as an alternative to subset selection
  - Overview: what regularization is & why it can help
  - Two main examples in linear regression
    - Ridge regression
    - The lasso (least absolute shrinkage and selection operator) – today and Fri

# Recap: Subset selection

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## Best subset selection:

- Exhaustively search all  $2^p$  subsets
- Pick the best model for each size  $k$ , then choose among them (via  $R_{\text{adj}}^2$  or CV)
- Feasible only for smaller  $p$  (high computational cost)

## Forward stepwise selection:

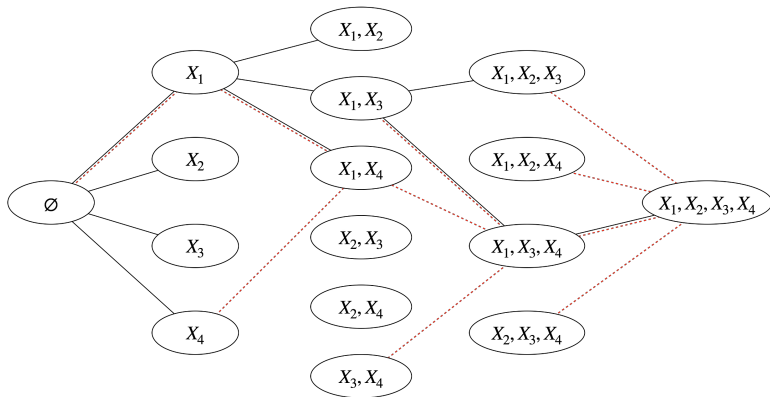
- Greedy approach: add one predictor at a time
- Complexity:  $\mathcal{O}(p^2)$  instead of  $2^p$
- May miss the global optimum if local decisions are suboptimal

## Backward stepwise selection:

- Greedy approach: remove one predictor at a time
- Similar pros/cons as forward stepwise

## Recap: Subset selection – Comparison of search paths

An illustrative example with  $p = 4$ :



**Figure:** Illustration of forward stepwise (**black solid path**; successively adding  $X_1 \rightarrow X_4 \rightarrow X_3 \rightarrow X_2$ ) and backward stepwise (**red dashed path**; removing  $X_2 \rightarrow X_3 \rightarrow X_4 \rightarrow X_1$ ). Best subset selection checks all  $2^p$  possibilities; all three can yield different outcomes.

## Recap: Subset selection – Summary

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- **Summary:**

- **Goal:** identify a relevant subset of predictors
- **Procedure:** evaluate subsets (all or partial), then pick best via  $R_{\text{adj}}^2$ , CV, etc.
- BSS is exhaustive but expensive; stepwise is faster
- Typically *refit the final chosen “best” subset using the standard least squares*

- **Advantages:**

- Direct variable selection: some  $\beta_j = 0$  (excluded)
- Straightforward implementation and intuitive interpretation

- **Disadvantages:**

- Even stepwise can be costly if  $p$  is very large
- Instability: small changes in the data can alter the chosen “best” subset

⇒ **Regularization** controls model complexity continuously by shrinking coefficients, often giving more stable or sparser estimates without searching over all subsets

# Regularization: What and why?

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**Recall least squares:** find parameters  $\hat{\beta}_0, \dots, \hat{\beta}_p$  that minimize

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \text{where} \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_{ip} x_{ip}$$

- **Challenges:**

- If  $p$  is large, predictors are highly correlated, or the data are noisy, least squares estimates can have high variance
- This can lead to overfitting, unstable coefficients, or no unique least-squares solution when  $p > n$

- **Idea:** *modify* the objective by **adding a penalty on  $\beta_j$ 's to stabilize fitting**

- Balance data fidelity vs. “simplicity” (by favoring smaller  $\beta_j$ )
- This approach is called *regularization* (or *shrinkage*)

We will learn two prominent regularization techniques for linear regression:

- **Ridge regression** ( $\ell_2$  penalty)
- **Lasso** ( $\ell_1$  penalty)

# Ridge regression: Formulation

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**Ridge regression:** Find  $\hat{\beta}_0, \dots, \hat{\beta}_p$  that minimize

$$\underbrace{\sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2}_{\text{RSS}} + \lambda \underbrace{\sum_{j=1}^p \beta_j^2}_{\text{penalty}}$$

- $\lambda \geq 0$  is a tuning parameter
- Each  $\lambda$  yields a different set of coefficient estimates  $\hat{\beta}_\lambda^R$

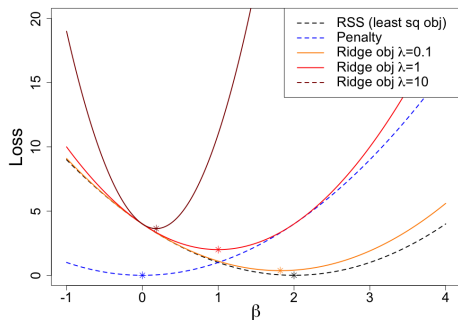
## Remarks:

- No penalty on  $\beta_0$  (the intercept)
- As  $\lambda \rightarrow 0$ , ridge regression  $\rightarrow$  standard least squares
- As  $\lambda$  grows,  $\beta_{1:p} = (\beta_1, \dots, \beta_p)$  shrinks toward 0
  - Reduces variance of  $\beta_{1:p}$  but increases bias

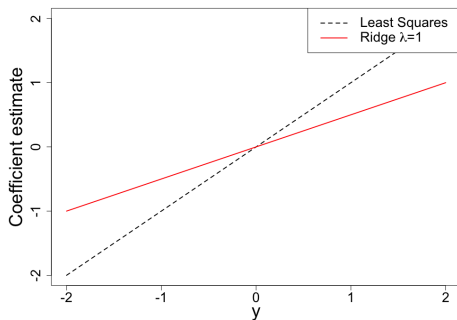
# Ridge regression: Illustration with 1D example

In the simplified setting with  $n = p = 1$  without intercept, ridge solves for  $\lambda \geq 0$ :

$$\hat{\beta}_{\lambda}^R \in \arg \min \left\{ (y - x\beta)^2 + \lambda\beta^2 \right\}$$



**Figure:** As  $\lambda$  grows,  $\hat{\beta}_{\lambda}^R$  shrinks toward 0 for fixed  $(y, x)$  ( $y = 2, x = 1$ ).



**Figure:** For each  $y$ ,  $\hat{\beta}_{\lambda}^R$  is smaller than the LS estimate  $y/x$  in magnitude, when  $\lambda > 0$ .

## Ridge regression: Illustration of shrinkage effects in 2D

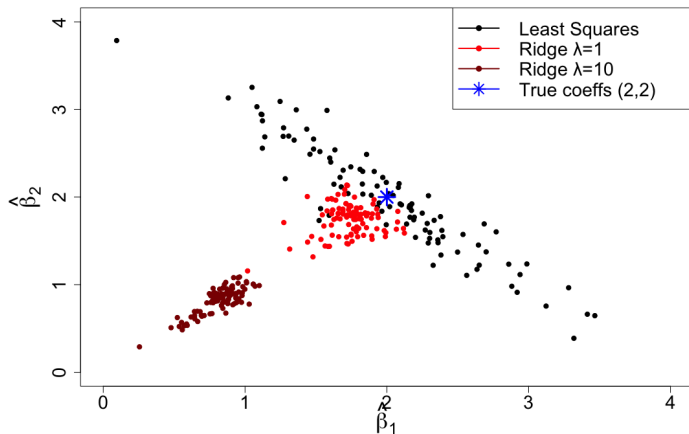


Figure: Scatter plots of 100 least squares estimates (**black**) vs. ridge estimates for  $\lambda = 1$  in **red** and  $\lambda = 10$  in **dark red**. As  $\lambda$  grows, the estimates cluster more tightly (lower variance) but shift away from the true value (**blue star**), indicating increased bias.

# Ridge regression: Effects of scaling variables & standardization

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The least squares coefficients are *scale equivariant*:

- Multiplying  $X_j$  by constant  $c$  scales  $\hat{\beta}_j$  by  $1/c$
- Regardless of scaling,  $X_j \hat{\beta}_j$  remains the same, not affecting other coefficients

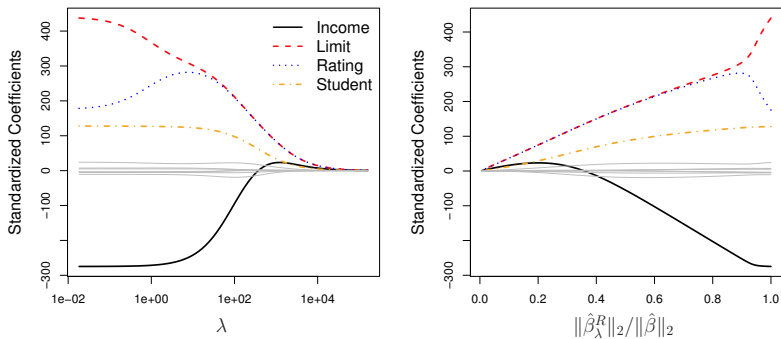
Ridge regression is *not* scale-equivariant:

- Rescaling one predictor can affect others through the penalty term
- Hence,  $X_j \hat{\beta}_{j,\lambda}^R$  depends on both  $\lambda$  *and* predictor scaling

Therefore, we usually *standardize predictors* before ridge:

$$\tilde{x}_{ij} = \frac{x_{ij}}{s_{ij}} \quad \text{where} \quad s_{ij}^2 = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 \quad \text{and} \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

## Ridge regression: Credit dataset example



**Figure:** Standardized ridge coefficients for **Credit**, plotted vs.  $\lambda$  and  $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$  [JWHT21, Figure 6.4].  
(Note:  $\|v\|_2 = \sqrt{v_1^2 + \dots + v_p^2}$ .)

- As  $\lambda$  increases, all  $\beta_j$  shrink toward 0 (none exactly zero though)
- If variables are correlated, ridge shrinks them together in a “group” manner

## Ridge regression: Advantages over least squares

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Ridge's advantage rooted in **Bias-variance tradeoff**:

- $\lambda = 0$ : no bias but high variance
- Larger  $\lambda$ : more bias but lower variance

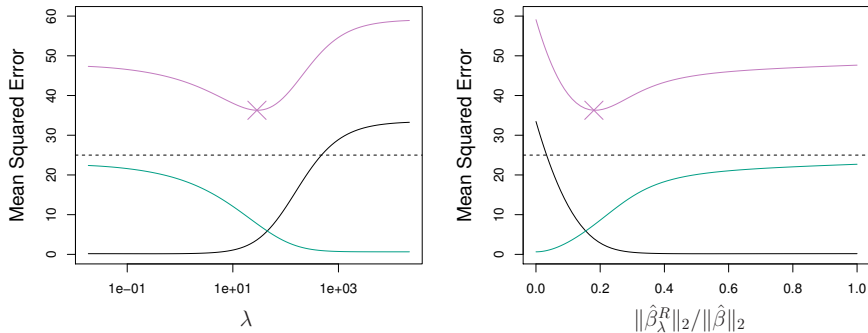
When ridge can be most helpful:

- If least squares has high variance (e.g.  $p \approx n$  or predictors are collinear)
- If data are noisy and  $\beta_j$  can fluctuate a lot
- For  $\lambda > 0$ , ridge still works even if  $p > n$ , producing a unique solution

Computational advantages of ridge:

- No need for  $2^p$  model fits (as in best subset); for any  $\lambda$ , ridge requires only a single fit
- Indeed, we can compute solutions for all  $\lambda$  at near the same cost as one OLS fit

# Ridge regression: Visualization of bias-variance tradeoff



**Figure:** Bias-variance tradeoff in ridge for a simulated data set ( $p = 45$ ,  $n = 50$ ). Shown are squared bias (**black**), variance (**green**), and test MSE (**purple**) vs.  $\lambda$  [JWHT21, Figure 6.5].

- At  $\lambda = 0$ , there is no bias but high variance
- Increasing  $\lambda$  *significantly* reduces variance at the cost of *slightly* higher bias
- Eventually, added bias overtakes the benefit of reduced variance

# Ridge regression: Example with a toy dataset (R script)

- **Problem setup:**
  - $n = 5, p = 2$
  - Ridge vs. least squares (at  $\lambda = 1$ )
- **Goal:** See how ridge regression shrinks coefficients  $\hat{\beta}$  compared to standard least squares

```
# Toy data: 5 obs, 2 predictors, no intercept
df <- data.frame(
  x1 = c(1,2,3,4,5),
  x2 = c(2,1,3,1,2),
  y  = c(2,2.5,6,4,6.5)
)
```

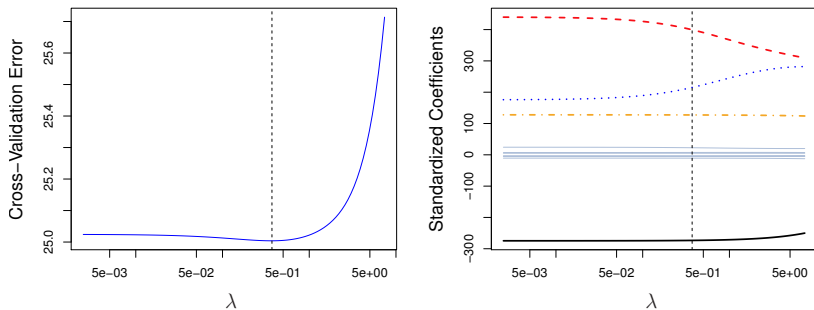
```
# OLS fit (no intercept => '-1')
ols_fit <- lm(y ~ x1 + x2 - 1, data = df)
cat("OLS Coeffs:\n", coef(ols_fit), "\n")

# If needed: install.packages("glmnet")
library(glmnet)

# Prepare X,y for glmnet
X <- as.matrix(df[,c("x1","x2")])
y <- df$y

# Ridge fit with lambda=1
ridge_fit <- glmnet(
  X, y, alpha=0, lambda=1,
  intercept=TRUE, standardize=TRUE
)
cat("Ridge Coeffs (lambda=1):\n", as.matrix(
  coef(ridge_fit)), "\n")
```

## Ridge regression: Selecting parameter $\lambda$



**Figure:** **Left:** CV errors for ridge regression on the **Credit** dataset with various values of  $\lambda$ . **Right:** Ridge regression coefficient estimates. The vertical dashed lines indicate the  $\lambda$  selected by CV [JWHT21, Figure 6.12].

- Each model with different  $\lambda$  is compared to each other
- Here, the chosen  $\lambda$  is relatively small, meaning minimal shrinkage relative to least square
- The error curve's dip is not very pronounced, suggesting a broad range of  $\lambda$  values yield similar performance

## Pop-up quiz: Ridge regression

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Which statement below is **false** about ridge regression?

(Hint: Think carefully. Which one might not always hold?)

- A) It often lowers variance compared to least squares, at the cost of higher bias.
- B) All ridge coefficients are strictly smaller in magnitude than the least squares estimates for each predictor.
- C) The tuning parameter  $\lambda$  strongly influences ridge performance, often chosen via cross-validation.
- D) Ridge tends to shrink correlated predictors together.

**Answer: B.**

**Brief explanation:**

- While ridge regression shrinks the overall magnitude of the coefficient vector, it does *not* guarantee every individual coefficient is always smaller than its least squares counterpart.
- In some cases, ridge can redistribute weights among correlated predictors, causing some estimates to exceed their OLS magnitudes even though the sum of squares  $\|\beta\|_2^2$  is reduced.

# Ridge regression: Summary

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- **Ridge regression formulation:**
  - Add a penalty term  $\sum_{j=1}^p \beta_j^2$  to the least squares objective, scaled by  $\lambda \geq 0$
  - Shrinks  $\beta_j$  more strongly as  $\lambda$  grows, stabilizing estimates
- **Bias-variance tradeoff:**
  - Larger  $\lambda \Rightarrow$  higher bias but lower variance
  - Especially helpful when OLS has high variance (e.g. large  $p$ , or  $p > n$ )
- **Computation:**
  - Efficient to solve for all  $\lambda$  at roughly the cost of one OLS fit
- **Limitation:**
  - Coefficients seldom reach exactly zero  $\implies$  no direct variable selection

# The lasso: Formulation

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**The lasso:** Find  $\hat{\beta}_0, \dots, \hat{\beta}_p$  that minimize

$$\underbrace{\sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2}_{\text{RSS}} + \underbrace{\lambda \sum_{j=1}^p |\beta_j|}_{\text{penalty}}$$

- $\lambda \geq 0$  is a tuning parameter
- Each choice of  $\lambda$  gives a different set of Lasso estimates  $\hat{\beta}_\lambda^L$

## Remarks:

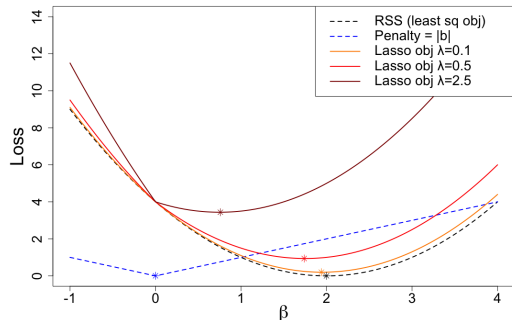
- No penalty on  $\beta_0$  (the intercept)
- As  $\lambda \rightarrow 0$ , lasso  $\rightarrow$  standard least squares
- As  $\lambda$  grows, many  $\beta_j$  shrink toward zero, and some exactly become 0

**Unlike ridge**, the lasso can yield exact zero estimates  $\implies$  **variable selection!**

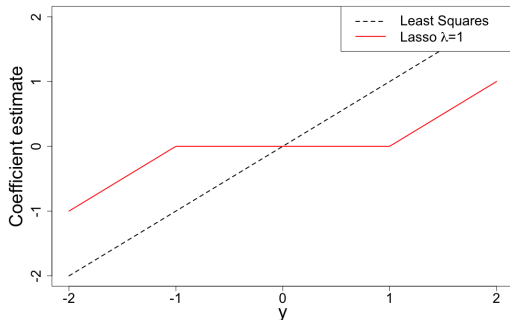
# Lasso: Illustration with 1D example

In the setting with  $n = p = 1$  without intercept, ridge solves for  $\lambda \geq 0$ :

$$\hat{\beta}_{\lambda}^R \in \arg \min \left\{ (y - x\beta)^2 + \lambda|\beta| \right\}$$



**Figure:** As  $\lambda$  grows,  $\hat{\beta}_{\lambda}^{\ell}$  shrinks more aggressively; small  $|y|$  can yield  $\beta = 0$  ( $y = 2$ ,  $x = 1$  fixed).



**Figure:** At  $\lambda = 1$ ,  $x = 1$ ,  $\beta$  hits 0 iff  $|y| \leq 1$ . This “thresholding” property underlies variable selection.

## Lasso: Regularization reduces variance, but...

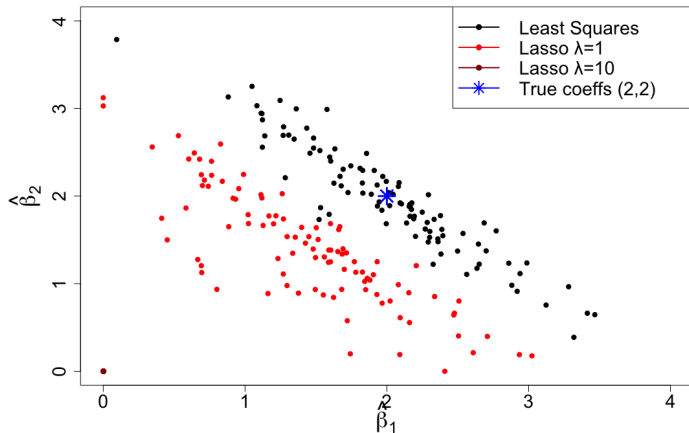


Figure: Scatter plots of 100 least squares estimates (**black**) vs. lasso estimates for  $\lambda = 1$  in **red** and  $\lambda = 10$  in **dark red**. Lasso can aggressively shrink or zero-out coefficients, but the variance reduction is less uniform than ridge. The shift from the true (**blue star**) may or may not be worth it.

## Lasso: Regularization enables variable selection

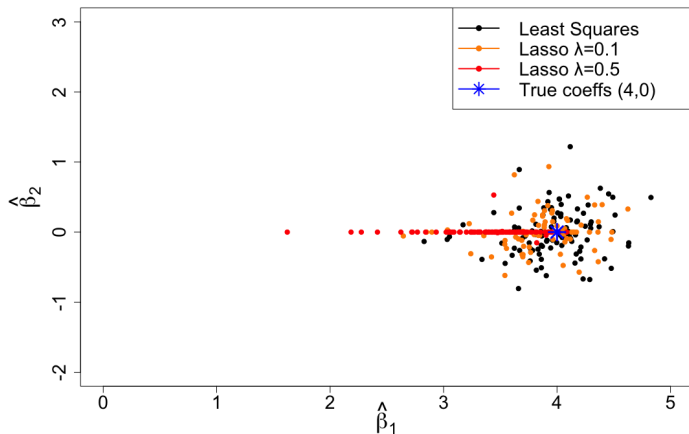


Figure: Scatter plots of 100 least squares estimates (black) vs. lasso estimates for  $\lambda = 0.1$  in orange and  $\lambda = 0.5$  in red. If the true  $\beta_2 = 0$  (blue star), lasso can correctly select the significant variable ( $X_1$ ), while suppressing noise and driving estimates to zero for  $X_2$ , thereby capturing the “sparse” true associations.

## The lasso: Credit dataset example

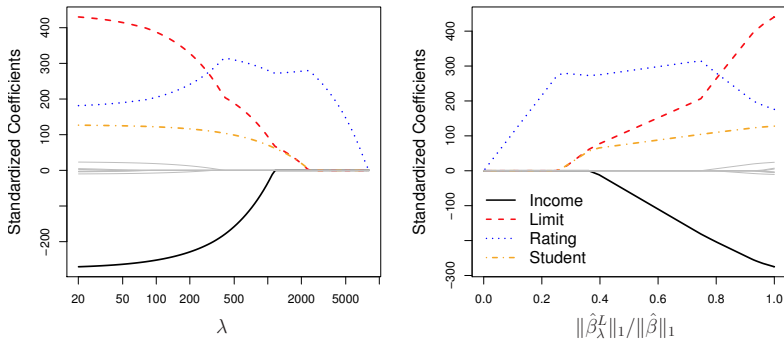


Figure: Standardized lasso coefficients for Credit, plotted vs.  $\lambda$  and the shrinkage level  $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$  [JWHT21, Figure 6.6].

- Lasso can force some coefficients to zero as  $\lambda$  increases
- Achieves *variable selection* directly (predictors with  $\hat{\beta}_j^L = 0$  are excluded)

# The lasso: Example with a toy dataset (R script)

- **Setup:**
  - $n = 5$ ,  $p = 2$
  - Compare lasso vs. OLS at  $\lambda = 1$
- **Objective:** See how lasso can shrink coefficients to zero

```
# Toy data: 5 obs, 2 predictors, no intercept
df <- data.frame(
  x1 = c(1,2,3,4,5),
  x2 = c(2,1,3,1,2),
  y  = c(2,2.5,6,4,6.5)
)
```

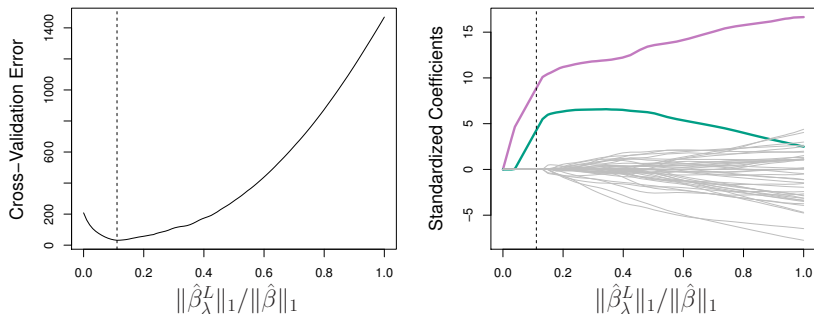
```
# OLS fit
ols_fit <- lm(y ~ x1 + x2, data=df)
cat("OLS Coeffs:\n", coef(ols_fit), "\n")

# If needed: install.packages("glmnet")
library(glmnet)

# Prepare X, y
X <- as.matrix(df[, c("x1","x2")])
y <- df$y

# Lasso with alpha=1, lambda=1
lasso_fit <- glmnet(X, y,
                    alpha=1,
                    lambda=1,
                    intercept=TRUE,
                    standardize=TRUE)
cat("Lasso Coeffs (lambda=1):\n",
    as.matrix(coef(lasso_fit)), "\n")
```

# The lasso: Selecting $\lambda$



**Figure:** **Left:** Ten-fold cross-validation MSE for the lasso, applied to the sparse simulated data. **Right:** The corresponding lasso coefficient estimates, with the two *signal* variables in color and the *noise* variables in gray. The vertical dashed line indicates the fit that minimizes the cross-validation error [JWHT21, Figure 6.13].

- The lasso cleanly separates two *signal* variables from *noise* variables
- In contrast, standard least squares (far right, with  $\|\hat{\beta}_\lambda^L\|_1 / \|\hat{\beta}\|_1 = 1$ ) only identifies the **purple** variable without discarding the noise predictors

# The lasso: Summary

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- **Lasso formulation:**
  - Penalty term  $\sum_{j=1}^p |\beta_j|$  scaled by  $\lambda$
  - As  $\lambda$  grows, some  $\beta_j$  become exactly zero  $\implies$  variable selection
- **Lasso advantages:**
  - Encourages a *sparse* model for interpretability
  - Computationally efficient in practice, though slightly more complex to solve than ridge
- **Ridge vs. lasso:**
  - Lasso can set some coefficients to *exact zeros*, while ridge generally does not
  - Ridge tends to be more stable especially when predictors are highly correlated
  - Both typically tune  $\lambda$  via cross-validation

## Wrap-up & Takeaways

### Regularization:

- **Ridge:**  $\ell_2$  penalty shrinks all coefficients toward zero, good if many have modest nonzero effects
- **The lasso:**  $\ell_1$  penalty can set some coefficients exactly to zero (variable selection)
- **Tuning**  $\lambda$  typically via cross-validation for both

|                    | Ridge                                          | Lasso                       |
|--------------------|------------------------------------------------|-----------------------------|
| Penalty            | $\sum_{j=1}^p \beta_j^2$                       | $\sum_{j=1}^p  \beta_j $    |
| Effect             | shrinks coefficients                           | shrinks and can set to zero |
| Variable selection | No                                             | Yes                         |
| Often useful when  | many small/moderate effects                    | only some predictors matter |
| Tuning             | $\lambda$ typically chosen by cross-validation |                             |

**Takeaway:** Ridge stabilizes estimates; lasso can produce sparse, interpretable models.

# References

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Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

*An Introduction to Statistical Learning: with Applications in R*, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.